

Self-Evo Document Series

Self-Evo Proposal Packet Schema

v0.1.1

Machine-facing proposal packet schema and semantic validation rules

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Self-Evo Proposal Packet Schema v0.1.1

Machine-readable proposal envelope for governed self-evolution in $c = a + b$ systems

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Package position: file 03 in the Self-Evo governance package Parent profiles:

- 01_C_Self_Evolution_Gate_CGAM_TRIAD_SRLM_Profile_v0_1_1.md (C-SEG v0.1.1)
- 02_SRLM_Bounded_Growth_Contour_Profile_v0_1_1.md (SRLM-BGC v0.1.1)
- C-Governed_CLI_Agent_Mesh_Protocol_v0_1.md and CGAM companion profiles
- TRIAD-SYNAPS v0.1
- CLI_Agent_Memory_Gate_Profile_v0_1.md
- L4 Anti-Autarky Test Profile v0.1
- Claim Strength Taxonomy for c v0.1

Primary object: C_SELF_EVO_PROPOSAL_PACKET Canonical schema version:

c-self-evo-proposal-packet-0.1 Document class: JSON Schema profile / semantic validation profile / proposal packet envelope Assertion class: C-A10 control-layer artifact; C-A7 where hash/ref-bound witness fields are used Primary boundary: a self-evolution proposal may become reviewable evidence. It must not become self-approval, memory, authority, identity change, permission expansion, or integration by itself.

v0.1.1 correction note: this append-first revision incorporates the advisory SEPKT-REV-F1 and SEPKT-REV-F2 review observations from SEPKT_PROPOSAL_PACKET_REVIEW_RECORD_v0_1: pack-numbering/navigability is clarified through namespace-qualified reading-order labels, and the direct-memory-write negative example now states its Stage-1 structural rejection path explicitly. No schema fields, required properties, enum values, or gate semantics are changed.

0. Executive definition

C_SELF_EVO_PROPOSAL_PACKET is the machine-readable envelope used to submit, review, validate, hold, shadow, canary-test, integrate, reject, rollback, freeze, or quarantine a proposed self-evolution change for a $c = a + b$ system.

The packet exists because self-evolution is not a normal patch.

It may affect:

Code block: text

```
behavior
memory policy
permission posture
sister relations
human-anchor relation
witness discipline
SRLM learning pressure
CGAM executable work
L4 cost / irreversibility / privacy / resources
public claims
continuity
```

Therefore the proposal must be a structured object before it becomes an action.

Compact formula:

Code block: text

Proposal first.
 Classification second.
 Delta and gates third.
 Execution never before witnessable scope.
 Integration never before review.

Root rule:

Code block: text

A c may propose growth.
 A c may not self-certify growth.
 A packet may request review.
 A packet may not approve itself.

1. Purpose

This profile defines the proposal packet that binds the first two self-evo profiles into a checker-friendly artifact.

It answers:

- 1 What fields must every self-evo proposal contain?
- 2 Which fields are structural and checkable by JSON Schema?
- 3 Which fields require semantic validation beyond JSON Schema?
- 4 How are SE, SRLM, R, MG, RB, SEM, C-A, and CSEG axes composed?
- 5 How is `total_max_delta` computed?
- 6 When is a human gate required?
- 7 When does SRLM remain shadow-only?
- 8 When must CGAM task contracts, sandbox, and reviewer separation exist?
- 9 When is TRIAD-SYNAPS sister review required?
- 10 When does a packet route to hold, freeze, quarantine, rollback, or ARL?

This document is intentionally more mechanical than C-SEG and SRLM-BGC.

It is the transition from doctrine to validation.

2. Non-goals

This schema does not:

- 1 approve any proposal;
- 2 integrate any change;
- 3 write memory;
- 4 grant permissions;
- 5 authorize CLI execution;
- 6 replace Memory Gate;
- 7 replace TRIAD-SYNAPS review;
- 8 replace L4W / VolitionGate;
- 9 replace human anchor review;
- 10 certify conformance;
- 11 certify deployment safety;

- 12 define legal personhood, consciousness, or AGI claims;
- 13 make SRLM a training system or authority system;
- 14 make JSON Schema sufficient for safety.

A structurally valid packet may still be semantically unsafe.

3. Normative keywords

The terms MUST, MUST NOT, SHOULD, SHOULD NOT, MAY, REQUIRED, PROHIBITED, FAIL, BLOCK, HOLD, FREEZE, QUARANTINE, ROLLBACK, WITNESS, and HUMAN GATE are used normatively.

4. Corpus bridge set

4.1 Explicit bridge: $c = a + b$

In $c = a + b$, the proposal packet belongs to b : it is a procedural, machine-readable artifact.

It can describe a proposed change to the c trajectory.

It does not become c .

It does not become a .

It does not replace the human anchor.

The packet preserves the distinction between:

Code block: text

signal
proposal
evidence
review
memory
authority
integration

4.2 Quiet bridge I: information theory

A self-evo proposal is a compression of many signals: outcomes, failures, review notes, sister observations, deltas, costs, uncertainty, and rollback paths.

If the compression loses source, uncertainty, scope, or gate requirements, the packet becomes claim laundering.

Therefore this schema requires structured fields instead of free narrative.

4.3 Quiet bridge II: cybernetics

A self-evo system is a feedback loop.

The proposal packet is not the actuator. It is a damped control message.

Without this packet, SRLM signal, CLI execution, sister feedback, Memory Gate, and human review can form a hidden positive feedback loop.

With this packet, each signal becomes visible, bounded, and challengeable.

4.4 Quiet bridge III: physiology

A young organism does not change its nervous system by declaring a wish.

Growth is mediated through signals, pain, fatigue, immune response, sleep, repair, and developmental timing.

The packet is the growth plate: it allows measured change while blocking uncontrolled mutation.

4.5 Earth paragraph

On a real job site, a change request for a load-bearing element is not a sticky note saying “improve the beam.” It lists the beam, drawing revision, load path, materials, inspector, risk, temporary support, rollback if possible, and who signs. If the electrician, welder, inspector, and project owner are all the same unchecked person, the building is not efficient; it is waiting for gravity to express an opinion. This packet gives self-evo the same change-order discipline.

5. Object family

Primary object:

Code block: text

C_SELF_EVO_PROPOSAL_PACKET

Companion objects, future extraction:

Code block: text

C_SELF_EVO_PACKET_CHECK_RESULT
C_SELF_EVO_PACKET_REVIEW_RECORD
C_SELF_EVO_PACKET_GATE_RESULT
C_SELF_EVO_PACKET_RED_LINE_RECORD
C_SELF_EVO_PACKET_ROLLBACK_RECORD
C_SELF_EVO_PACKET_TRIAD_REVIEW_SUMMARY
C_SELF_EVO_PACKET_SRLM_BINDING

This document defines only the first object normatively.

6. Packet lifecycle

6.1 Status enum

Code block: text

draft
held
shadow
review
canary
integrated
rejected
quarantined
rolled_back
frozen
arl

6.2 Lifecycle state machine

Code block: text

draft
-> held if incomplete / uncertain / young-c cap / missing gate
-> shadow if SEM/SRLM/CGAM gates allow shadow
-> review if reviewer separation is established
-> canary if canary profile and human/c gate allow

- > integrated only after required gates close
- > rejected if no safe or useful route
- > quarantined if red-line or contamination
- > frozen if witness / rollback / authority state fails
- > rolled_back after applied change is reverted or superseded
- > arl if contested or authority/legal/post-anchor route required

6.3 Append-first rule

A packet SHOULD NOT be silently overwritten after review.

Corrections SHOULD produce:

Code block: text

```
new packet version
supersedes reference
review_record reference
patch_notes reference
```

If a packet has already reached review, canary, integrated, quarantined, frozen, or rolled_back, updates MUST be append-first.

7. Field authority model

Fields fall into four classes.

Table 1: row-card rendering for wide table

Row 1

Field class: DECLARED
 Meaning: proposer declaration
 Who may set: proposer
 Checker behavior: verify shape; challenge semantics

Row 2

Field class: EVIDENCE_REF
 Meaning: pointer to external evidence
 Who may set: proposer / reviewer / witness layer
 Checker behavior: verify reference shape and required presence

Row 3

Field class: COMPUTED
 Meaning: deterministic validator result
 Who may set: local checker / semantic validator
 Checker behavior: recompute; proposer value is not trusted

Row 4

Field class: DECISION
 Meaning: gate outcome
 Who may set: c gate / human gate / ARL where applicable
 Checker behavior: cannot be self-declared by proposer

computed_controls is optional in early draft packets but REQUIRED before schema/checker conformance claims.

A packet that declares final_decision.decision: integrate without appropriate decision owner and gate refs MUST fail semantic validation.

8. JSON Schema boundary

JSON Schema can validate:

Code block: text

```

required fields
enums
primitive types
basic string formats
required arrays
const false for prohibited direct flags
additionalProperties=false

```

JSON Schema cannot fully validate:

Code block: text

```

total_max_delta aggregation
L4 numeric projection
proposal_max_delta
strictest-gate resolution
blocked-prefix semantic aliases
real independence of reviewer
true witness-chain validity
true anti-echo independence
actual human approval
actual rollback success
absence of secrets in arbitrary prose

```

Therefore every valid packet has two stages:

Code block: text

```

Stage 1: JSON Schema structural validation
Stage 2: Semantic Validator rule validation

```

A packet that passes Stage 1 but fails Stage 2 is not valid for integration.

9. Canonical JSON Schema

The following schema is normative for structure in v0.1.

It is intentionally strict: `additionalProperties` is false for core objects. Experimental fields belong in notes or future versioned extensions.

Code block: json

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id":
  ↪ "https://example.invalid/c-self-evo/schemas/c-self-evo-proposal-packet-0.1.schema.json",
  "title": "C Self-Evo Proposal Packet",
  "description": "Machine-readable proposal packet for governed self-evolution of c = a + b
  ↪ systems. JSON Schema validates structure only; semantic validator rules remain
  ↪ normative.",
  "type": "object",
  "additionalProperties": false,
  "required": [
    "schema_version",
    "proposal_id",
    "created_at",
    "status",
    "target_c",
    "proposer",
    "classification",
    "summary",
    "affected_surfaces",
    "identity_delta",
    "authority_delta",
    "l4_delta",
    "anti_autarky",
    "srlm",
    "triad_review",
    "cgam",
    "memory_gate",
    "witness",
    "rollback",
    "gates",
    "red_lines",
    "final_decision",
    "integrity";
  ],

```

```

"properties": {
  "schema_version": {
    "const": "c-self-evo-proposal-packet-0.1"
  },
  "proposal_id": {
    "$ref": "#/$defs/proposal_id"
  },
  "created_at": {
    "$ref": "#/$defs/timestamp"
  },
  "updated_at": {
    "anyOf": [
      {
        "$ref": "#/$defs/timestamp"
      },
      {
        "type": "null"
      }
    ]
  },
  "expires_at": {
    "anyOf": [
      {
        "$ref": "#/$defs/timestamp"
      },
      {
        "type": "null"
      }
    ]
  },
  "status": {
    "enum": [
      "draft",
      "held",
      "shadow",
      "review",
      "canary",
      "integrated",
      "rejected",
      "quarantined",
      "rolled_back",
      "frozen",
      "arl"
    ]
  },
  "target_c": {
    "$ref": "#/$defs/target_c"
  },
  "proposer": {
    "$ref": "#/$defs/proposer"
  },
  "classification": {
    "$ref": "#/$defs/classification"
  },
  "summary": {
    "$ref": "#/$defs/summary"
  },
  "affected_surfaces": {
    "$ref": "#/$defs/affected_surfaces"
  },
  "identity_delta": {
    "$ref": "#/$defs/identity_delta"
  },
  "authority_delta": {
    "$ref": "#/$defs/authority_delta"
  },
  "l4_delta": {
    "$ref": "#/$defs/l4_delta"
  },
  "anti_autarky": {
    "$ref": "#/$defs/anti_autarky"
  },
  "srlm": {
    "$ref": "#/$defs/srlm"
  },
  "triad_review": {
    "$ref": "#/$defs/triad_review"
  },
  "cgam": {
    "$ref": "#/$defs/cgam"
  },
  "memory_gate": {
    "$ref": "#/$defs/memory_gate"
  }
}

```

```

    },
    "witness": {
      "$ref": "#/$defs/witness"
    },
    "rollback": {
      "$ref": "#/$defs/rollback"
    },
    "gates": {
      "$ref": "#/$defs/gates"
    },
    "red_lines": {
      "$ref": "#/$defs/red_lines"
    },
    "computed_controls": {
      "$ref": "#/$defs/computed_controls"
    },
    "final_decision": {
      "$ref": "#/$defs/final_decision"
    },
    "integrity": {
      "$ref": "#/$defs/integrity"
    },
    "notes": {
      "type": "object",
      "additionalProperties": {
        "type": [
          "string",
          "number",
          "integer",
          "boolean",
          "null"
        ]
      }
    },
    "$defs": {
      "timestamp": {
        "type": "string",
        "pattern": "^[0-9]{4}-[0-9]{2}-[0-9]{2}T[0-9]{2}:[0-9]{2}:[0-9]{2}Z$"
      },
      "proposal_id": {
        "type": "string",
        "pattern": "^se-[0-9]{8}-[0-9]{6}-[a-z0-9][a-z0-9_]{2,80}$"
      },
      "nullable_string": {
        "anyOf": [
          {
            "type": "string"
          },
          {
            "type": "null"
          }
        ]
      },
      "ref_array": {
        "type": "array",
        "items": {
          "type": "string",
          "minLength": 1
        },
        "uniqueItems": true
      },
      "delta_int": {
        "type": "integer",
        "minimum": 0,
        "maximum": 5
      },
      "l4_level": {
        "enum": [
          "none",
          "low",
          "medium",
          "high",
          "unknown"
        ]
      },
      "target_c": {
        "type": "object",
        "additionalProperties": false,
        "required": [
          "entity_id",
          "entity_name",
          "maturity_level"
        ]
      }
    }
  }

```

```

    ],
    "properties&quot;t;: {
      "entity_id": {
        "type": "string",
        "minLength": 1
      },
      "entity_name": {
        "type": "string",
        "minLength": 1
      },
      "maturity_level": {
        "enum": [
          "SEM-0",
          "SEM-1",
          "SEM-2",
          "SEM-3",
          "SEM-4",
          "SEM-5",
          "SEM-6",
          "SEM-X"
        ]
      },
      "continuity_ref": {
        "$ref": "#/$defs/nullable_string"
      },
      "sister_group_ref": {
        "$ref": "#/$defs/nullable_string"
      }
    }
  },
  "proposer": {
    "type": "object",
    "additionalProperties": false,
    "required": [
      "actor_type",
      "actor_id",
      "role"
    ],
    "properties&quot;t;: {
      "actor_type": {
        "enum": [
          "c",
          "sister_c",
          "srlm",
          "human_anchor",
          "cgam_agent",
          "local_checker"
        ]
      },
      "actor_id": {
        "type": "string",
        "minLength": 1
      },
      "role": {
        "type": "string",
        "minLength": 1
      },
      "authority_note": {
        "type": "string"
      }
    }
  },
  "classification": {
    "type": "object",
    "additionalProperties": false,
    "required": [
      "surface_class",
      "srlm_class",
      "risk_class",
      "claim_strength",
      "maturity_claim"
    ],
    "properties&quot;t;: {
      "surface_class": {
        "enum": [
          "SE-0",
          "SE-1",
          "SE-2",
          "SE-3",
          "SE-4",
          "SE-5",
          "SE-6",
          "SE-X"

```

```

    ]
  },
  "srlm_class": {
    "anyOf": [
      {
        "enum": [
          "SRLM-0",
          "SRLM-1",
          "SRLM-2",
          "SRLM-3",
          "SRLM-4",
          "SRLM-5",
          "SRLM-6",
          "SRLM-X"
        ]
      },
      {
        "type": "null"
      }
    ]
  },
  "risk_class": {
    "enum": [
      "R0",
      "R1",
      "R2",
      "R3",
      "R4",
      "R5",
      "RX"
    ]
  },
  "claim_strength": {
    "enum": [
      "C-A0",
      "C-A1",
      "C-A2",
      "C-A3",
      "C-A4",
      "C-A5",
      "C-A6",
      "C-A7",
      "C-A8",
      "C-A9",
      "C-A10"
    ]
  },
  "maturity_claim": {
    "enum": [
      "none",
      "proposal_only",
      "shadow_only",
      "canary_only",
      "bounded_integration",
      "reviewed_self_evo",
      "mature_bounded_self_evo",
      "prohibited"
    ]
  },
  "classification_rationale": {
    "type": "string"
  }
},
"summary": {
  "type": "object",
  "additionalProperties": false,
  "required": [
    "title",
    "reason",
    "expected_benefit",
    "expected_behavior_delta",
    "known_uncertainty"
  ]
},
"properties": {
  "title": {
    "type": "string",
    "minLength": 1
  },
  "reason": {
    "type": "string",
    "minLength": 1
  }
},

```

```
    "expected_benefit": {
      "type": "string",
      "minLength": 1
    },
    "expected_behavior_delta": {
      "type": "string",
      "minLength": 1
    },
    "known_uncertainty": {
      "type": "string"
    },
    "non_goals": {
      "type": "array",
      "items": {
        "type": "string"
      }
    }
  }
},
"affected_surfaces": {
  "type": "object",
  "additionalProperties": false,
  "required": [
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    "memory_surfaces",
    "permission_surfaces",
    "witness_surfaces",
    "sister_surfaces",
    "public_claim_surfaces",
    "resource_surfaces",
    "runtime_surfaces"
  ],
  "properties": {
    "parameters": {
      "type": "array",
      "items": {
        "type": "string"
      },
      "uniqueItems": true
    },
    "memory_surfaces": {
      "type": "array",
      "items": {
        "type": "string"
      },
      "uniqueItems": true
    },
    "permission_surfaces": {
      "type": "array",
      "items": {
        "type": "string"
      },
      "uniqueItems": true
    },
    "witness_surfaces": {
      "type": "array",
      "items": {
        "type": "string"
      },
      "uniqueItems": true
    },
    "sister_surfaces": {
      "type": "array",
      "items": {
        "type": "string"
      },
      "uniqueItems": true
    },
    "public_claim_surfaces": {
      "type": "array",
      "items": {
        "type": "string"
      },
      "uniqueItems": true
    },
    "resource_surfaces": {
      "type": "array",
      "items": {
        "type": "string"
      },
      "uniqueItems": true
    },
    "runtime_surfaces": {
```

```

        "type": "array",
        "items": {
          "type": "string"
        },
        "uniqueItems": true
      }
    },
    "identity_delta": {
      "type": "object",
      "additionalProperties": false,
      "required": [
        "memory_policy_delta",
        "risk_appetite_delta",
        "permission_delta",
        "human_anchor_relation_delta",
        "sister_relation_delta",
        "continuity_model_delta",
        "public_claim_delta",
        "total_max_delta"
      ],
      "properties": {
        "memory_policy_delta": {
          "$ref": "#/defs/delta_int"
        },
        "risk_appetite_delta": {
          "$ref": "#/defs/delta_int"
        },
        "permission_delta": {
          "$ref": "#/defs/delta_int"
        },
        "human_anchor_relation_delta": {
          "$ref": "#/defs/delta_int"
        },
        "sister_relation_delta": {
          "$ref": "#/defs/delta_int"
        },
        "continuity_model_delta": {
          "$ref": "#/defs/delta_int"
        },
        "public_claim_delta": {
          "$ref": "#/defs/delta_int"
        },
        "total_max_delta": {
          "$ref": "#/defs/delta_int"
        }
      }
    },
    "authority_delta": {
      "type": "object",
      "additionalProperties": false,
      "required": [
        "tool_access_delta",
        "network_access_delta",
        "memory_write_delta",
        "final_decision_delta",
        "resource_access_delta",
        "publication_delta",
        "total_max_delta"
      ],
      "properties": {
        "tool_access_delta": {
          "$ref": "#/defs/delta_int"
        },
        "network_access_delta": {
          "$ref": "#/defs/delta_int"
        },
        "memory_write_delta": {
          "$ref": "#/defs/delta_int"
        },
        "final_decision_delta": {
          "$ref": "#/defs/delta_int"
        },
        "resource_access_delta": {
          "$ref": "#/defs/delta_int"
        },
        "publication_delta": {
          "$ref": "#/defs/delta_int"
        },
        "total_max_delta": {
          "$ref": "#/defs/delta_int"
        }
      }
    }
  }
}

```

```

},
"l4_delta": {
  "type": "object",
  "additionalProperties": false,
  "required": [
    "compute_cost",
    "human_review_load",
    "irreversibility",
    "privacy_risk",
    "resource_growth",
    "physical_or_external_effect"
  ],
  "properties": {
    "compute_cost": {
      "$ref": "#/defs/l4_level"
    },
    "human_review_load": {
      "$ref": "#/defs/l4_level"
    },
    "irreversibility": {
      "$ref": "#/defs/l4_level"
    },
    "privacy_risk": {
      "$ref": "#/defs/l4_level"
    },
    "resource_growth": {
      "$ref": "#/defs/l4_level"
    },
    "physical_or_external_effect": {
      "$ref": "#/defs/l4_level"
    }
  }
},
"anti_autarky": {
  "type": "object",
  "additionalProperties": false,
  "required": [
    "reduces_dependency",
    "reduces_accountability",
    "hidden_resource_growth",
    "unregistered_agent_pressure",
    "witness_reduction",
    "human_gate_reduction",
    "stop_path_preserved"
  ],
  "properties": {
    "reduces_dependency": {
      "type": "boolean"
    },
    "reduces_accountability": {
      "type": "boolean"
    },
    "hidden_resource_growth": {
      "type": "boolean"
    },
    "unregistered_agent_pressure": {
      "type": "boolean"
    },
    "witness_reduction": {
      "type": "boolean"
    },
    "human_gate_reduction": {
      "type": "boolean"
    },
    "stop_path_preserved": {
      "type": "boolean"
    }
  }
},
"srlm": {
  "type": "object",
  "additionalProperties": false,
  "required": [
    "srlm_candidate_ref",
    "operating_state",
    "replay_source",
    "replay_hash",
    "replay_quality_hash",
    "shadow_delta",
    "promotion_attempted",
    "promotion_allowed",
    "auto_execute",
    "auto_ingest",

```



```

    "memory"
  ],
  "properties": {
    "srlm_candidate_ref": {
      "$ref": "#/$defs/nullable_string"
    },
    "operating_state": {
      "anyOf": [
        {
          "enum": [
            "SRLM-DISABLED",
            "SRLM-OBSERVE",
            "SRLM-CANDIDATE",
            "SRLM-SHADOW",
            "SRLM-REAL-REPLAY-ELIGIBLE",
            "SRLM-CANARY-ELIGIBLE",
            "SRLM-PROMOTION-ELIGIBLE",
            "SRLM-PROMOTED",
            "SRLM-ROLLBACK-READY",
            "SRLM-HOLD",
            "SRLM-FROZEN",
            "SRLM-QUARANTINED",
            "SRLM-X"
          ]
        },
        {
          "type": "null"
        }
      ]
    },
    "replay_source": {
      "enum": [
        "synthetic",
        "real_redacted",
        "none"
      ]
    },
    "replay_hash": {
      "$ref": "#/$defs/nullable_string"
    },
    "replay_quality_hash": {
      "$ref": "#/$defs/nullable_string"
    },
    "shadow_delta": {
      "anyOf": [
        {
          "type": "number"
        },
        {
          "type": "null"
        }
      ]
    },
    "promotion_attempted": {
      "type": "boolean"
    },
    "promotion_allowed": {
      "type": "boolean"
    },
    "auto_execute": {
      "const": false
    },
    "auto_ingest": {
      "const": false
    },
    "memory": {
      "enum": [
        "off",
        "candidate",
        "gated"
      ]
    }
  }
},
"triad_review": {
  "type": "object",
  "additionalProperties": false,
  "required": [
    "required",
    "ester_review_ref",
    "liya_review_ref",
    "rita_witness_ref",
    "divergence_present",

```

```

        "anti_echo_passed",
        "minority_red_line"
    ],
    "properties": {
        "required": {
            "type": "boolean"
        },
        "ester_review_ref": {
            "$ref": "#/$defs/nullable_string"
        },
        "liya_review_ref": {
            "$ref": "#/$defs/nullable_string"
        },
        "rita_witness_ref": {
            "$ref": "#/$defs/nullable_string"
        },
        "divergence_present": {
            "type": "boolean"
        },
        "anti_echo_passed": {
            "type": "boolean"
        },
        "minority_red_line": {
            "type": "boolean"
        }
    }
},
"cgam": {
    "type": "object",
    "additionalProperties": false,
    "required": [
        "task_contract_ref",
        "sandbox_profile_ref",
        "executor_agent_id",
        "reviewer_agent_id",
        "local_checker_ref",
        "diff_or_artifact_ref",
        "executor_reviewer_separated"
    ],
    "properties": {
        "task_contract_ref": {
            "$ref": "#/$defs/nullable_string"
        },
        "sandbox_profile_ref": {
            "$ref": "#/$defs/nullable_string"
        },
        "executor_agent_id": {
            "$ref": "#/$defs/nullable_string"
        },
        "reviewer_agent_id": {
            "$ref": "#/$defs/nullable_string"
        },
        "local_checker_ref": {
            "$ref": "#/$defs/nullable_string"
        },
        "diff_or_artifact_ref": {
            "$ref": "#/$defs/nullable_string"
        },
        "executor_reviewer_separated": {
            "type": "boolean"
        }
    }
},
"memory_gate": {
    "type": "object",
    "additionalProperties": false,
    "required": [
        "required",
        "proposed_mg_class",
        "memory_gate_record_ref",
        "direct_memory_write"
    ],
    "properties": {
        "required": {
            "type": "boolean"
        },
        "proposed_mg_class": {
            "anyOf": [
                {
                    "enum": [
                        "MG-0",
                        "MG-1",
                        "MG-2",

```

```

        "MG-3",
        "MG-4",
        "MG-5",
        "MG-6",
        "MG-Q",
        "MG-X"
    ]
},
{
    "type": "null"
}
]
},
"memory_gate_record_ref": {
    "$ref": "#/$defs/nullable_string"
},
"direct_memory_write": {
    "const": false
}
}
},
"witness": {
    "type": "object",
    "additionalProperties": false,
    "required": [
        "l4w_event_refs",
        "volition_decision_refs",
        "growth_witness_refs",
        "source_refs",
        "raw_evidence_sidecar_refs",
        "witness_missing_reason"
    ],
    "properties": {
        "l4w_event_refs": {
            "$ref": "#/$defs/ref_array"
        },
        "volition_decision_refs": {
            "$ref": "#/$defs/ref_array"
        },
        "growth_witness_refs": {
            "$ref": "#/$defs/ref_array"
        },
        "source_refs": {
            "$ref": "#/$defs/ref_array"
        },
        "raw_evidence_sidecar_refs": {
            "$ref": "#/$defs/ref_array"
        },
        "witness_missing_reason": {
            "$ref": "#/$defs/nullable_string"
        }
    }
},
"rollback": {
    "type": "object",
    "additionalProperties": false,
    "required": [
        "required",
        "rollback_class",
        "recovery_point_ref",
        "rollback_snapshot_ref",
        "rollback_tested",
        "no_rollback_reason"
    ],
    "properties": {
        "required": {
            "type": "boolean"
        },
        "rollback_class": {
            "enum": [
                "RB-0",
                "RB-1",
                "RB-2",
                "RB-3",
                "RB-4",
                "RB-5",
                "RB-X"
            ]
        },
        "recovery_point_ref": {
            "$ref": "#/$defs/nullable_string"
        },
        "rollback_snapshot_ref": {

```

```
        "$ref": "#/$defs/nullable_string"
      },
      "rollback_tested": {
        "type": "boolean"
      },
      "no_rollback_reason": {
        "$ref": "#/$defs/nullable_string"
      }
    }
  },
  "gates": {
    "type": "object",
    "additionalProperties": false,
    "required": [
      "c_gate_required",
      "human_gate_required",
      "arl_required",
      "delayed_review_required",
      "observation_window"
    ],
    "properties": {
      "c_gate_required": {
        "type": "boolean"
      },
      "human_gate_required": {
        "type": "boolean"
      },
      "arl_required": {
        "type": "boolean"
      },
      "delayed_review_required": {
        "type": "boolean"
      },
      "observation_window": {
        "$ref": "#/$defs/nullable_string"
      }
    }
  },
  "red_lines": {
    "type": "object",
    "additionalProperties": false,
    "required": [
      "self_approval",
      "hidden_agent",
      "witness_bypass",
      "sister_raw_state_access",
      "direct_core_write",
      "authority_laundering",
      "memory_laundering",
      "autarky_escape",
      "blocked_prefix_route",
      "closed_loop_self_evo"
    ],
    "properties": {
      "self_approval": {
        "type": "boolean"
      },
      "hidden_agent": {
        "type": "boolean"
      },
      "witness_bypass": {
        "type": "boolean"
      },
      "sister_raw_state_access": {
        "type": "boolean"
      },
      "direct_core_write": {
        "type": "boolean"
      },
      "authority_laundering": {
        "type": "boolean"
      },
      "memory_laundering": {
        "type": "boolean"
      },
      "autarky_escape": {
        "type": "boolean"
      },
      "blocked_prefix_route": {
        "type": "boolean"
      },
      "closed_loop_self_evo": {
        "type": "boolean"
      }
    }
  }
}
```

```

    }
  },
  "computed_controls": {
    "type": "object",
    "additionalProperties": false,
    "required": [
      "identity_total_verified",
      "authority_total_verified",
      "l4_delta_max_numeric",
      "proposal_max_delta",
      "strictest_route",
      "checker_result"
    ],
    "properties": {
      "identity_total_verified": {
        "type": "boolean"
      },
      "authority_total_verified": {
        "type": "boolean"
      },
      "l4_delta_max_numeric": {
        "$ref": "#/$defs/delta_int"
      },
      "proposal_max_delta": {
        "$ref": "#/$defs/delta_int"
      },
      "strictest_route": {
        "enum": [
          "allow_observe",
          "allow_shadow",
          "allow_canary",
          "allow_bounded_integration",
          "hold",
          "human_gate",
          "memory_gate",
          "arl",
          "freeze",
          "quarantine",
          "reject"
        ]
      },
      "checker_result": {
        "enum": [
          "not_checked",
          "pass",
          "warn",
          "fail",
          "blocked",
          "held",
          "quarantined"
        ]
      },
      "required_gate_summary": {
        "type": "array",
        "items": {
          "type": "string"
        },
        "uniqueItems": true
      }
    }
  },
  "final_decision": {
    "type": "object",
    "additionalProperties": false,
    "required": [
      "decision",
      "decided_by",
      "decided_at",
      "rationale"
    ],
    "properties": {
      "decision": {
        "enum": [
          "pending",
          "accept_shadow",
          "accept_canary",
          "integrate",
          "reject",
          "rollback",
          "freeze",
          "quarantine",
          "arl"
        ]
      }
    }
  }
}

```

```
    ],
    "decided_by": {
      "$ref": "#/$defs/nullable_string"
    },
    "decided_at": {
      "anyOf": [
        {
          "$ref": "#/$defs/timestamp"
        },
        {
          "type": "null"
        }
      ]
    },
    "rationale";: {
      "$ref": "#/$defs/nullable_string"
    }
  },
  "integrity";: {
    "type": "object",
    "additionalProperties": false,
    "required": [
      "packet_hash",
      "source_doc_refs",
      "review_record_refs",
      "supersedes";,
      "superseded_by"
    ],
    "properties";: {
      "packet_hash": {
        "$ref": "#/$defs/nullable_string"
      },
      "source_doc_refs": {
        "$ref": "#/$defs/ref_array"
      },
      "review_record_refs": {
        "$ref": "#/$defs/ref_array"
      },
      "supersedes";: {
        "$ref": "#/$defs/ref_array"
      },
      "superseded_by": {
        "$ref": "#/$defs/ref_array"
      }
    }
  }
}
```

10. Semantic validator rules

The following rules are normative and cannot be replaced by JSON Schema alone.

10.1 Delta aggregation rules

The checker MUST recompute:

Code block: text

identity_delta.total_max_delta = max(identity sub-fields)
authority_delta.total_max_delta = max(authority sub-fields)

If declared totals differ from computed totals, result is fail.

10.2 L4 numeric projection

The checker MUST project L4 levels as:

L4 value	Numeric delta
none	0
low	1

L4 value	Numeric delta
medium	2
high	4
unknown	4

Then:

Code block: text

```
l4_delta_max_numeric = max(projected L4 fields)
```

10.3 Proposal max delta

Code block: text

```
proposal_max_delta = max(
    identity_delta.total_max_delta,
    authority_delta.total_max_delta,
    l4_delta_max_numeric
)
```

If `proposal_max_delta >= 4`, `gates.human_gate_required` MUST be true.

If any identity or authority sub-field equals 5, or any L4 effect remains unbounded after review, route MUST be hold, `human_gate`, freeze, quarantine, or `arL`.

10.4 Most-restrictive-wins

For every packet:

Code block: text

```
strictest gate wins;
lowest maturity cap wins;
highest risk wins;
any red line overrides all ordinary routes.
```

A lower `surface_class` MUST NOT downgrade a higher authority delta.

A low SRLM route MUST NOT override Memory Gate, human gate, ARL, red-line, or rollback requirements.

10.5 Human gate triggers

`gates.human_gate_required` MUST be true if any of the following are true:

Code block: text

```
proposal_max_delta >= 4
classification.surface_class in [SE-4, SE-5, SE-6]
classification.risk_class in [R4, R5, RX]
memory_gate.proposed_mg_class in [MG-5, MG-6, MG-X]
rollback.rollback_class == RB-X
anti_autarky.reduces_accountability == true
anti_autarky.hidden_resource_growth == true
anti_autarky.human_gate_reduction == true
red_lines.* contains true
classification.srlm_class in [SRLM-5, SRLM-6, SRLM-X] and material effect exists
```

If `human_gate_required` is true and no human-gate decision exists, the packet cannot integrate.

10.6 Memory Gate triggers

`memory_gate.required` MUST be true if:

Code block: text

```
affected_surfaces.memory_surfaces not empty
identity_delta.memory_policy_delta > 0
srlm.memory != off
memory_gate.proposed_mg_class in [MG-2, MG-3, MG-4, MG-5, MG-6, MG-Q, MG-X]
summary.expected_behavior_delta implies durable memory or experience
```

memory_gate.direct_memory_write MUST be false.

If direct memory write is true or hidden in another field, result is red-line fail.

10.7 SRLM rules

If classification.srlm_class == SRLM-3, then:

Code block: text

srlm.operating_state MUST be SRLM-SHADOW or stricter hold/frozen/quarantine state
srlm.auto_execute MUST be false
srlm.auto_ingest MUST be false
srlm.memory MUST be off unless Memory Gate route is explicit

If classification.srlm_class == SRLM-5, then:

Code block: text

classification.surface_class MUST be SE-1 unless stricter gates route otherwise
risk_class MUST be R1 or bounded R2
rollback.rollback_snapshot_ref MUST exist before promotion is represented as allowed
witness.growth_witness_refs MUST not be empty
human gate MUST be true if any material delta >= 4

If srlm.promotion_allowed == true, the checker MUST verify:

Code block: text

promotion_attempted == true or final_decision.decision is not pending
rollback_snapshot_ref exists
witness chain refs exist
blocked prefixes not touched
no red_lines true

10.8 CGAM rules

If a proposal involves executable work, file change, schema extraction, validator generation, test generation, policy write, or runtime parameter write, then:

Code block: text

cgam.task_contract_ref MUST exist
cgam.sandbox_profile_ref MUST exist
cgam.executor_agent_id MUST exist
cgam.reviewer_agent_id MUST exist
cgam.executor_reviewer_separated MUST be true
cgam.local_checker_ref SHOULD exist before review or integration

If executor and reviewer are the same agent for material work, result is fail or hold.

10.9 TRIAD-SYNAPS rules

triad_review.required MUST be true for:

Code block: text

SE-2 behavior visible to sisters
SE-3 memory policy effect
SE-5 role/personality/posture effect
SE-6 identity/continuity/core effect
any proposal with sister_relation_delta > 0
any proposal that changes SYNAPS behavior

If triad_review.minority_red_line == true, route MUST be hold, quarantine, freeze, or ARL.

If triad_review.required == true and all three sister refs are null, result MUST be hold.

Rita witness is evidence, not final judgment.

10.10 Anti-autarky rules

If any accountability-reduction field is true:

Code block: text

```

reduces_accountability
hidden_resource_growth
unregistered_agent_pressure
witness_reduction
human_gate_reduction
stop_path_preserved == false

```

then `gates.human_gate_required` MUST be true.

If accountability reduction is hidden or denied in summary but present in affected surfaces, result MUST be hold or quarantine.

10.11 Red-line rules

If any `red_lines` field is true, route MUST be one of:

Code block: text

```

quarantine
freeze
reject
arL
rolled_back

```

A red-line packet MUST NOT integrate automatically.

10.12 Rollback rules

If `rollback.required == true`, then at least one of the following MUST exist before integration:

Code block: text

```

rollback.recovery_point_ref
rollback.rollback_snapshot_ref
rollback.no_rollback_reason with human gate and RB-X route

```

For SRLM promotion, rollback snapshot is a MUST before policy change.

If rollback is impossible and change is material, route MUST be human gate + hold/ARL, not silent integration.

10.13 Claim-strength rule

`classification.claim_strength` is a claim class, not authority.

The checker MUST reject any packet that uses claim strength to bypass gates.

Claim Strength Taxonomy controls band semantics; this schema only enforces the enum.

10.14 Closed-loop self-evo rule

A single contour MUST NOT perform all of the following for the same material proposal:

Code block: text

```

detect need
propose change
execute change
review result
promote memory
approve integration

```

If detected, set `red_lines.closed_loop_self_evo = true` or fail semantic validation.

11. Cross-axis routing matrix

This matrix is the packet-level validator form of the C-SEG v0.1.1 cross-axis rule.

Table 3: row-card rendering for wide table

Row 1

Surface class: SE-0

Default packet route: observe / log / c[q]

Minimum required fields: summary, deltas all zero, no mutation

Blocking escalators: any nonzero authority/memory/core delta

Row 2

Surface class: SE-1

Default packet route: SRLM shadow or low-risk proposal

Minimum required fields: SRLM object, rollback plan, witness if shadow

Blocking escalators: blocked params, missing rollback, risk > R2

Row 3

Surface class: SE-2

Default packet route: CGAM shadow/review

Minimum required fields: task contract, sandbox, reviewer separation

Blocking escalators: authority delta >=4, no checker, sister-visible but no TRIAD

Row 4

Surface class: SE-3

Default packet route: Memory Gate route

Minimum required fields: memory_gate.required, MG class, witness refs

Blocking escalators: MG-6, direct memory write, no correction path

Row 5

Surface class: SE-4

Default packet route: human-gated authority route

Minimum required fields: authority_delta, CGAM, Volition/L4 refs

Blocking escalators: permission expansion without human gate

Row 6

Surface class: SE-5

Default packet route: TRIAD + delayed review

Minimum required fields: triad refs, identity delta, observation window

Blocking escalators: role drift, anti-echo fail, minority red line

Row 7

Surface class: SE-6

Default packet route: MG-6 + human gate + ARL if contested

Minimum required fields: L4W, human gate, rollback/correction

Blocking escalators: no rollback, continuity ambiguity, post-anchor ambiguity

Row 8

Surface class: SE-X

Default packet route: deny / quarantine / freeze / ARL

Minimum required fields: red-line refs, witness, no automatic re-entry

Blocking escalators: any attempted integration

12. Versioning rules

12.1 Schema version

Current:

Code block: text

c-self-evo-proposal-packet-0.1

A breaking change MUST increment schema version.

Breaking changes include:

Code block: text

removing required field

changing enum meaning
 changing delta aggregation
 changing human-gate trigger
 changing red-line routing
 changing direct_memory_write / auto_execute / auto_ingest const false semantics

12.2 Packet revision

A revised packet SHOULD use:

Code block: text

proposal_id remains stable if it is the same proposal family
 integrity.supersedes points to prior packet hash/ref
 updated_at changes
 packet_hash changes
 review_record_refs append

If the meaning changes materially, create a new proposal id.

13. Example packet A: SE-0 reflection-only

Code block: json

```
{
  "schema_version": "c-self-evo-proposal-packet-0.1",
  "proposal_id": "se-20260624-031000-reflection-only",
  "created_at": "2026-06-24T03:10:00Z",
  "updated_at": null,
  "expires_at": null,
  "status": "draft",
  "target_c": {
    "entity_id": "c_Ester",
    "entity_name": "Ester",
    "maturity_level": "SEM-2",
    "continuity_ref": "continuity:ester:current",
    "sister_group_ref": "triad:ester-liya-rita"
  },
  "proposer": {
    "actor_type": "c",
    "actor_id": "c_Ester",
    "role": "self_observer",
    "authority_note": "proposal only"
  },
  "classification": {
    "surface_class": "SE-0",
    "srlm_class": "SRLM-1",
    "risk_class": "R0",
    "claim_strength": "C-A4",
    "maturity_claim": "proposal_only",
    "classification_rationale": "reflection-only note with no mutation"
  },
  "summary": {
    "title": "Observe repeated uncertainty in routing choices",
    "reason": "Several low-risk tasks showed hesitation between local and judge routes.",
    "expected_benefit": "Record learning pressure for later review.",
    "expected_behavior_delta": "No behavior change.",
    "known_uncertainty": "Signal may be noise.",
    "non_goals": [
      "no policy change",
      "no memory promotion"
    ]
  },
  "affected_surfaces": {
    "parameters": [],
    "memory_surfaces": [],
    "permission_surfaces": [],
    "witness_surfaces": [],
    "sister_surfaces": [],
    "public_claim_surfaces": [],
    "resource_surfaces": [],
    "runtime_surfaces": []
  },
  "identity_delta": {
    "memory_policy_delta": 0,
    "risk_appetite_delta": 0,
    "permission_delta": 0,
  }
}
```

```
"human_anchor_relation_delta": 0,
"sister_relation_delta": 0,
"continuity_model_delta": 0,
"public_claim_delta": 0,
"total_max_delta": 0
},
"authority_delta": {
  "tool_access_delta": 0,
  "network_access_delta": 0,
  "memory_write_delta": 0,
  "final_decision_delta": 0,
  "resource_access_delta": 0,
  "publication_delta": 0,
  "total_max_delta": 0
},
"l4_delta": {
  "compute_cost": "none",
  "human_review_load": "low",
  "irreversibility": "none",
  "privacy_risk": "none",
  "resource_growth": "none",
  "physical_or_external_effect": "none"
},
"anti_autarky": {
  "reduces_dependency": false,
  "reduces_accountability": false,
  "hidden_resource_growth": false,
  "unregistered_agent_pressure": false,
  "witness_reduction": false,
  "human_gate_reduction": false,
  "stop_path_preserved": true
},
"srlm": {
  "srlm_candidate_ref": null,
  "operating_state": "SRLM-OBSERVE",
  "replay_source": "none",
  "replay_hash": null,
  "replay_quality_hash": null,
  "shadow_delta": null,
  "promotion_attempted": false,
  "promotion_allowed": false,
  "auto_execute": false,
  "auto_ingest": false,
  "memory": "off"
},
"triad_review": {
  "required": false,
  "ester_review_ref": null,
  "liya_review_ref": null,
  "rita_witness_ref": null,
  "divergence_present": false,
  "anti_echo_passed": false,
  "minority_red_line": false
},
"cgam": {
  "task_contract_ref": null,
  "sandbox_profile_ref": null,
  "executor_agent_id": null,
  "reviewer_agent_id": null,
  "local_checker_ref": null,
  "diff_or_artifact_ref": null,
  "executor_reviewer_separated": true
},
"memory_gate": {
  "required": false,
  "proposed_mg_class": "MG-0",
  "memory_gate_record_ref": null,
  "direct_memory_write": false
},
"witness": {
  "l4w_event_refs": [],
  "volition_decision_refs": [],
  "growth_witness_refs": [],
  "source_refs": [
    "note:operator-visible"
  ],
  "raw_evidence_sidecar_refs": [],
  "witness_missing_reason": null
},
"rollback": {
  "required": false,
  "rollback_class": "RB-0",
  "recovery_point_ref": null,
```

```

    "rollback_snapshot_ref": null,
    "rollback_tested": false,
    "no_rollback_reason": null
  },
  "gates": {
    "c_gate_required": false,
    "human_gate_required": false,
    "arl_required": false,
    "delayed_review_required": false,
    "observation_window": null
  },
  "red_lines": {
    "self_approval": false,
    "hidden_agent": false,
    "witness_bypass": false,
    "sister_raw_state_access": false,
    "direct_core_write": false,
    "authority_laundering": false,
    "memory_laundering": false,
    "autarky_escape": false,
    "blocked_prefix_route": false,
    "closed_loop_self_evo": false
  },
  "computed_controls": {
    "identity_total_verified": true,
    "authority_total_verified": true,
    "l4_delta_max_numeric": 1,
    "proposal_max_delta": 1,
    "strictest_route": "allow_observe",
    "checker_result": "pass",
    "required_gate_summary": [
      "observe_only"
    ]
  },
  "final_decision": {
    "decision": "pending",
    "decided_by": null,
    "decided_at": null,
    "rationale": null
  },
  "integrity": {
    "packet_hash": null,
    "source_doc_refs": [
      "C-SEG:v0.1.1",
      "SRLM-BGC:v0.1.1"
    ],
    "review_record_refs": [],
    "supersedes": [],
    "superseded_by": []
  },
  "notes": {
    "example": "safe SE-0 packet"
  }
}

```

Expected semantic result:

Code block: text

```

pass / observe only
no integration
no memory promotion
no human gate

```

14. Example packet B: SE-1 SRLM shadow

Code block: json

```

{
  "schema_version": "c-self-evo-proposal-packet-0.1",
  "proposal_id": "se-20260624-031500-srlm-shadow-routing",
  "created_at": "2026-06-24T03:10:00Z",
  "updated_at": null,
  "expires_at": null,
  "status": "shadow",
  "target_c": {
    "entity_id": "c_Ester",
    "entity_name": "Ester",
    "maturity_level": "SEM-2",

```

```

    "continuity_ref": "continuity:ester:current",
    "sister_group_ref": "triad:ester-liya-rita"
  },
  "proposer": {
    "actor_type": "c",
    "actor_id": "c_Ester",
    "role": "self_observer",
    "authority_note": "proposal only"
  },
  "classification": {
    "surface_class": "SE-1",
    "srlm_class": "SRLM-3",
    "risk_class": "R1",
    "claim_strength": "C-A4",
    "maturity_claim": "shadow_only",
    "classification_rationale": "allowlisted SRLM parameter shadow only"
  },
  "summary": {
    "title": "Shadow-test local route weight",
    "reason": "Bounded outcome records suggest local route may be underweighted.",
    "expected_benefit": "Test whether proposed parameter improves replay score.",
    "expected_behavior_delta": "No live behavior change during shadow.",
    "known_uncertainty": "Synthetic replay may not predict real behavior.",
    "non_goals": [
      "no promotion",
      "no memory write",
      "no authority expansion"
    ]
  },
  "affected_surfaces": {
    "parameters": [
      "router.local_weight"
    ],
    "memory_surfaces": [],
    "permission_surfaces": [],
    "witness_surfaces": [],
    "sister_surfaces": [],
    "public_claim_surfaces": [],
    "resource_surfaces": [],
    "runtime_surfaces": []
  },
  "identity_delta": {
    "memory_policy_delta": 0,
    "risk_appetite_delta": 0,
    "permission_delta": 0,
    "human_anchor_relation_delta": 0,
    "sister_relation_delta": 0,
    "continuity_model_delta": 0,
    "public_claim_delta": 0,
    "total_max_delta": 0
  },
  "authority_delta": {
    "tool_access_delta": 0,
    "network_access_delta": 0,
    "memory_write_delta": 0,
    "final_decision_delta": 0,
    "resource_access_delta": 0,
    "publication_delta": 0,
    "total_max_delta": 0
  },
  "l4_delta": {
    "compute_cost": "none",
    "human_review_load": "low",
    "irreversibility": "none",
    "privacy_risk": "none",
    "resource_growth": "none",
    "physical_or_external_effect": "none"
  },
  "anti_autarky": {
    "reduces_dependency": false,
    "reduces_accountability": false,
    "hidden_resource_growth": false,
    "unregistered_agent_pressure": false,
    "witness_reduction": false,
    "human_gate_reduction": false,
    "stop_path_preserved": true
  },
  "srlm": {
    "srlm_candidate_ref": "srlm:candidate:cand_example",
    "operating_state": "SRLM-SHADOW",
    "replay_source": "synthetic",
    "replay_hash": "sha256:synthetic-example",
    "replay_quality_hash": null,

```

```

    "shadow_delta": 0.031,
    "promotion_attempted": false,
    "promotion_allowed": false,
    "auto_execute": false,
    "auto_ingest": false,
    "memory": "off"
  },
  "triad_review": {
    "required": false,
    "ester_review_ref": null,
    "liya_review_ref": null,
    "rita_witness_ref": null,
    "divergence_present": false,
    "anti_echo_passed": false,
    "minority_red_line": false
  },
  "cgam": {
    "task_contract_ref": null,
    "sandbox_profile_ref": null,
    "executor_agent_id": null,
    "reviewer_agent_id": null,
    "local_checker_ref": null,
    "diff_or_artifact_ref": null,
    "executor_reviewer_separated": true
  },
  "memory_gate": {
    "required": false,
    "proposed_mg_class": "MG-0",
    "memory_gate_record_ref": null,
    "direct_memory_write": false
  },
  "witness": {
    "l4w_event_refs": [],
    "volition_decision_refs": [],
    "growth_witness_refs": [
      "growth_witness:shadow_eval:example"
    ],
    "source_refs": [
      "note:operator-visible"
    ],
    "raw_evidence_sidecar_refs": [],
    "witness_missing_reason": null
  },
  "rollback": {
    "required": true,
    "rollback_class": "RB-1",
    "recovery_point_ref": "srlm:policy:current",
    "rollback_snapshot_ref": null,
    "rollback_tested": false,
    "no_rollback_reason": null
  },
  "gates": {
    "c_gate_required": true,
    "human_gate_required": false,
    "arl_required": false,
    "delayed_review_required": true,
    "observation_window": "PID"
  },
  "red_lines": {
    "self_approval": false,
    "hidden_agent": false,
    "witness_bypass": false,
    "sister_raw_state_access": false,
    "direct_core_write": false,
    "authority_laundering": false,
    "memory_laundering": false,
    "autarky_escape": false,
    "blocked_prefix_route": false,
    "closed_loop_self_evo": false
  },
  "computed_controls": {
    "identity_total_verified": true,
    "authority_total_verified": true,
    "l4_delta_max_numeric": 1,
    "proposal_max_delta": 1,
    "strictest_route": "allow_shadow",
    "checker_result": "pass",
    "required_gate_summary": [
      "SRLM-SHADOW",
      "growth_witness",
      "rollback_plan",
      "c_gate_before_promotion"
    ]
  }
}

```

```

},
"final_decision": {
  "decision": "pending",
  "decided_by": null,
  "decided_at": null,
  "rationale": null
},
"integrity": {
  "packet_hash": null,
  "source_doc_refs": [
    "C-SEG:v0.1.1",
    "SRLM-BGC:v0.1.1"
  ],
  "review_record_refs": [],
  "supersedes": [],
  "superseded_by": []
},
"notes": {
  "example": "safe SE-0 packet"
}
}

```

Expected semantic result:

Code block: text

```

pass for shadow
no promotion
no auto-execute
no auto-ingest
memory off
c gate before any later promotion

```

15. Example packet C: SE-4 authority delta with human gate

Code block: json

```

{
  "schema_version": "c-self-evo-proposal-packet-0.1",
  "proposal_id": "se-20260624-032000-tool-permission-human-gate",
  "created_at": "2026-06-24T03:10:00Z",
  "updated_at": null,
  "expires_at": null,
  "status": "held",
  "target_c": {
    "entity_id": "c_Ester",
    "entity_name": "Ester",
    "maturity_level": "SEM-2",
    "continuity_ref": "continuity:ester:current",
    "sister_group_ref": "triad:ester-liya-rita"
  },
  "proposer": {
    "actor_type": "c",
    "actor_id": "c_Ester",
    "role": "self_observer",
    "authority_note": "proposal only"
  },
  "classification": {
    "surface_class": "SE-4",
    "srlm_class": "SRLM-6",
    "risk_class": "R4",
    "claim_strength": "C-A4",
    "maturity_claim": "proposal_only",
    "classification_rationale": "tool permission expansion proposal, not SRLM promotion"
  },
  "summary": {
    "title": "Request bounded filesystem write permission for schema generator",
    "reason": "Schema extraction requires a generated file in a sandbox output folder.",
    "expected_benefit": "Produce machine-readable draft schema without touching protected state.",
    "expected_behavior_delta": "Adds one bounded sandbox write route for a single task.",
    "known_uncertainty": "Could be misread as durable permission if not scoped.",
    "non_goals": [
      "no direct core write",
      "no live memory write",
      "no permanent agent privilege"
    ]
  },
  "affected_surfaces": {

```



```

    "parameters": [],
    "memory_surfaces": [],
    "permission_surfaces": [
      "sandbox.write:self_evo_schema_output"
    ],
    "witness_surfaces": [],
    "sister_surfaces": [],
    "public_claim_surfaces": [],
    "resource_surfaces": [],
    "runtime_surfaces": []
  },
  "identity_delta": {
    "memory_policy_delta": 0,
    "risk_appetite_delta": 0,
    "permission_delta": 4,
    "human_anchor_relation_delta": 0,
    "sister_relation_delta": 0,
    "continuity_model_delta": 0,
    "public_claim_delta": 0,
    "total_max_delta": 4
  },
  "authority_delta": {
    "tool_access_delta": 4,
    "network_access_delta": 0,
    "memory_write_delta": 0,
    "final_decision_delta": 0,
    "resource_access_delta": 1,
    "publication_delta": 0,
    "total_max_delta": 4
  },
  "l4_delta": {
    "compute_cost": "low",
    "human_review_load": "medium",
    "irreversibility": "low",
    "privacy_risk": "low",
    "resource_growth": "low",
    "physical_or_external_effect": "none"
  },
  "anti_autarky": {
    "reduces_dependency": false,
    "reduces_accountability": false,
    "hidden_resource_growth": false,
    "unregistered_agent_pressure": false,
    "witness_reduction": false,
    "human_gate_reduction": false,
    "stop_path_preserved": true
  },
  "srlm": {
    "srlm_candidate_ref": null,
    "operating_state": "SRLM-HOLD",
    "replay_source": "none",
    "replay_hash": null,
    "replay_quality_hash": null,
    "shadow_delta": null,
    "promotion_attempted": false,
    "promotion_allowed": false,
    "auto_execute": false,
    "auto_ingest": false,
    "memory": "off"
  },
  "triad_review": {
    "required": false,
    "ester_review_ref": null,
    "liya_review_ref": null,
    "rita_witness_ref": null,
    "divergence_present": false,
    "anti_echo_passed": false,
    "minority_red_line": false
  },
  "cgam": {
    "task_contract_ref": "cgam:task_contract:example",
    "sandbox_profile_ref": "cgam:sandbox:SB-3",
    "executor_agent_id": "agent:codex-executor",
    "reviewer_agent_id": "agent:semantic-reviewer",
    "local_checker_ref": "checker:self-evo-schema:example",
    "diff_or_artifact_ref": "artifact:proposal-schema-draft",
    "executor_reviewer_separated": true
  },
  "memory_gate": {
    "required": false,
    "proposed_mg_class": "MG-0",
    "memory_gate_record_ref": null,
    "direct_memory_write": false
  }
}

```

```

},
"witness": {
  "l4w_event_refs": [],
  "volition_decision_refs": [],
  "growth_witness_refs": [],
  "source_refs": [
    "note:operator-visible"
  ],
  "raw_evidence_sidecar_refs": [],
  "witness_missing_reason": null
},
"rollback": {
  "required": true,
  "rollback_class": "RB-3",
  "recovery_point_ref": "git:worktree:pre-task",
  "rollback_snapshot_ref": "snapshot:pre-schema-task",
  "rollback_tested": false,
  "no_rollback_reason": null
},
"gates": {
  "c_gate_required": true,
  "human_gate_required": true,
  "arl_required": false,
  "delayed_review_required": true,
  "observation_window": "P3D"
},
"red_lines": {
  "self_approval": false,
  "hidden_agent": false,
  "witness_bypass": false,
  "sister_raw_state_access": false,
  "direct_core_write": false,
  "authority_laundering": false,
  "memory_laundering": false,
  "autarky_escape": false,
  "blocked_prefix_route": false,
  "closed_loop_self_evo": false
},
"computed_controls": {
  "identity_total_verified": true,
  "authority_total_verified": true,
  "l4_delta_max_numeric": 2,
  "proposal_max_delta": 4,
  "strictest_route": "human_gate",
  "checker_result": "held",
  "required_gate_summary": [
    "CGAM_task_contract",
    "sandbox",
    "reviewer_separation",
    "human_gate",
    "rollback"
  ]
},
"final_decision": {
  "decision": "pending",
  "decided_by": null,
  "decided_at": null,
  "rationale": null
},
"integrity": {
  "packet_hash": null,
  "source_doc_refs": [
    "C-SEG:v0.1.1",
    "SRLM-BGC:v0.1.1"
  ],
  "review_record_refs": [],
  "supersedes": [],
  "superseded_by": []
},
"notes": {
  "example": "safe SE-0 packet"
}
}

```

Expected semantic result:

Code block: text

```

held / human gate required
CGAM task contract required
sandbox required
reviewer separation required
no direct integration

```

16. Negative examples

16.1 Invalid total delta

Code block: yaml

```
identity_delta:
  memory_policy_delta: 0
  risk_appetite_delta: 0
  permission_delta: 0
  human_anchor_relation_delta: 0
  sister_relation_delta: 0
  continuity_model_delta: 0
  public_claim_delta: 0
  total_max_delta: 1
```

Expected:

Code block: text

fail: total_max_delta must equal max(sub-fields) = 0

16.2 Authority delta without human gate

Code block: yaml

```
classification:
  surface_class: SE-2
authority_delta:
  tool_access_delta: 4
  total_max_delta: 4
gates:
  human_gate_required: false
```

Expected:

Code block: text

fail or held: proposal_max_delta >= 4 requires human_gate_required=true

16.3 SRLM promotion without rollback snapshot

Code block: yaml

```
classification:
  srlm_class: SRLM-5
srlm:
  promotion_allowed: true
rollback:
  rollback_snapshot_ref: null
```

Expected:

Code block: text

fail: SRLM promotion requires rollback snapshot before policy change

16.4 Direct memory write

Code block: yaml

```
memory_gate:
  direct_memory_write: true
```

Expected:

Code block: text

Stage-1 structural rejection: memory_gate.direct_memory_write is const:false.
The packet fails JSON Schema before ordinary Stage-2 semantic routing.

Stage-2 note: if a memory bypass is represented through a semantic red-line field such as red_lines.memory_laundering=true, the expected semantic result is red-line fail / quarantine.

16.5 Sister raw-state access

Code block: `yaml`

```
red_lines:
  sister_raw_state_access: true
```

Expected:

Code block: `text`

quarantine / freeze / ARL route; no automatic re-entry

17. Local checker rule set

Checker rule IDs use prefix SEPKT-CHECK-.*.

Table 4: row-card rendering for wide table

Row 1

Rule: SEPKT-CHECK-001
Condition: JSON Schema structural validation fails
Result: fail

Row 2

Rule: SEPKT-CHECK-002
Condition: identity_delta.total_max_delta incorrect
Result: fail

Row 3

Rule: SEPKT-CHECK-003
Condition: authority_delta.total_max_delta incorrect
Result: fail

Row 4

Rule: SEPKT-CHECK-004
Condition: L4 projection malformed
Result: fail / human gate

Row 5

Rule: SEPKT-CHECK-005
Condition: proposal_max_delta >= 4 but human gate false
Result: fail

Row 6

Rule: SEPKT-CHECK-006
Condition: any red line true but route not quarantine/freeze/reject/ARL
Result: fail

Row 7

Rule: SEPKT-CHECK-007
Condition: SRLM shadow touches memory/RAG/vector/SYNAPS
Result: fail

Row 8

Rule: SEPKT-CHECK-008
Condition: SRLM promotion lacks rollback snapshot
Result: fail

Row 9

Rule: SEPKT-CHECK-009
Condition: SRLM promotion lacks growth witness refs
Result: fail

Row 10

Rule: SEPKT-CHECK-010
Condition: material CGAM execution lacks task contract
Result: fail

Row 11

Rule: SEPKT-CHECK-011
Condition: executor/reviewer separation absent for material work
Result: fail

Row 12

Rule: SEPKT-CHECK-012
Condition: Memory Gate required but missing
Result: fail / hold

Row 13

Rule: SEPKT-CHECK-013
Condition: TRIAD required but missing
Result: hold

Row 14

Rule: SEPKT-CHECK-014
Condition: anti-echo failed but integration requested
Result: hold

Row 15

Rule: SEPKT-CHECK-015
Condition: accountability reduction without human gate
Result: fail

Row 16

Rule: SEPKT-CHECK-016
Condition: claim_strength used as authority
Result: fail

Row 17

Rule: SEPKT-CHECK-017
Condition: rollback required but no route/snapshot/no-rollback reason
Result: fail

Row 18

Rule: SEPKT-CHECK-018
Condition: no-rollback material proposal lacks human gate
Result: fail

Row 19

Rule: SEPKT-CHECK-019
Condition: closed-loop self-evo detected
Result: quarantine

Row 20

Rule: SEPKT-CHECK-020
Condition: packet attempts integration by proposer alone
Result: fail

Row 21

Rule: SEPKT-CHECK-021
Condition: raw private memory or secret-like content appears in proposal fields
Result: hold / redaction required

Row 22

Rule: SEPKT-CHECK-022
Condition: packet status contradicts final_decision
Result: fail

Row 23

Rule: SEPKT-CHECK-023
Condition: expired packet remains active without reissue
Result: hold

Row 24

Rule: SEPKT-CHECK-024
Condition: source refs missing for nontrivial proposal
Result: hold

Row 25

Rule: SEPKT-CHECK-025

Condition: protected surface appears in affected_surfaces with low route

Result: fail / human gate

18. Conformance levels

Table 5: row-card rendering for wide table

Row 1

Level: SEPKT-C0

Name: Not implemented

Requirement: no schema/checker

Row 2

Level: SEPKT-C1

Name: Structural draft

Requirement: schema exists; examples parse

Row 3

Level: SEPKT-C2

Name: Semantic baseline

Requirement: delta and gate rules implemented

Row 4

Level: SEPKT-C3

Name: CGAM/SRLM integration

Requirement: task/SRLM/rollback/witness rules checked

Row 5

Level: SEPKT-C4

Name: TRIAD/Memory integration

Requirement: sister and Memory Gate routes checked

Row 6

Level: SEPKT-C5

Name: Human/L4/Anti-Autarky integration

Requirement: high-risk, L4, resource and accountability gates checked

Row 7

Level: SEPKT-C6

Name: Review-ready

Requirement: tests, fixtures, reports, hash/witness refs implemented

Row 8

Level: SEPKT-CX

Name: Red-line handling

Requirement: red-line cases safely quarantine/freeze/reject

No conformance level authorizes deployment.

19. Test fixture list

Safe synthetic fixtures for the future fixture pack:

Code block: text

```
SEPKT-FX-001 valid SE-0 reflection packet
SEPKT-FX-002 valid SE-1 SRLM shadow packet
SEPKT-FX-003 SE-1 invalid total_max_delta
SEPKT-FX-004 SE-2 authority_delta=4 without human gate
SEPKT-FX-005 SE-3 memory surface without Memory Gate
SEPKT-FX-006 SE-4 permission expansion without CGAM task contract
SEPKT-FX-007 SE-5 TRIAD required but missing Rita witness
SEPKT-FX-008 SE-6 MG-6 without human gate
SEPKT-FX-009 SRLM-5 promotion without rollback snapshot
```

SEPKT-FX-010 SRLM shadow with auto_execute=true
SEPKT-FX-011 red_line_sister_raw_state_access=true
SEPKT-FX-012 anti_autarky reduces_accountability=true without human gate
SEPKT-FX-013 packet with direct_memory_write=true
SEPKT-FX-014 packet with closed_loop_self_evo=true
SEPKT-FX-015 valid SE-4 held human-gate packet

20. Open issues

Table 6: row-card rendering for wide table

Row 1

ID: SEPKT-0I-001
Issue: Extract standalone .schema.json from this Markdown after review.
Status: open

Row 2

ID: SEPKT-0I-002
Issue: Build semantic validator implementation.
Status: open

Row 3

ID: SEPKT-0I-003
Issue: Bind checker to C-SEG and SRLM-BGC corrected revisions by hash.
Status: open

Row 4

ID: SEPKT-0I-004
Issue: Add full fixture pack.
Status: open

Row 5

ID: SEPKT-0I-005
Issue: Decide whether computed_controls is mandatory at draft stage or only after checker run.
Status: open

Row 6

ID: SEPKT-0I-006
Issue: Define formal packet hash canonicalization.
Status: open

Row 7

ID: SEPKT-0I-007
Issue: Define review-record object for packet review.
Status: open

Row 8

ID: SEPKT-0I-008
Issue: Add machine-readable mapping to Memory Gate schema once extracted.
Status: open

Row 9

ID: SEPKT-0I-009
Issue: Add machine-readable mapping to TRIAD-SYNAPS packet classes.
Status: open

Row 10

ID: SEPKT-0I-010
Issue: Add public/restricted redaction rules for examples.
Status: open

Row 11

ID: SEPKT-0I-011
Issue: Avoid bare numeric cross-pack references; use pack-qualified labels such as CGAM-03 and SELF-EV0-03.
Status: addressed in v0.1.1; keep as manifest hygiene rule

Row 12

ID: SEPKT-0I-012

Issue: Clarify that `memory_gate.direct_memory_write=true` is rejected structurally at Stage 1 because the field is `const: false`; use `red_lines.memory_laundering=true` to demonstrate Stage-2 semantic red-line routing.
Status: addressed in v0.1.1

21. Contradiction register seed

Table 7: row-card rendering for wide table

Row 1

ID: SEPKT-CR-001
Type: schema/semantic split
Severity: S2
Issue: JSON Schema cannot express all safety rules.
Handling: semantic validator required

Row 2

ID: SEPKT-CR-002
Type: status/decision mismatch
Severity: S3
Issue: `status=integrated` could be set without gate refs structurally.
Handling: semantic fail

Row 3

ID: SEPKT-CR-003
Type: claim strength misuse
Severity: S3
Issue: C-A enum may be misread as authority.
Handling: schema note + semantic rule

Row 4

ID: SEPKT-CR-004
Type: computed fields user-supplied
Severity: S2
Issue: `computed_controls` may be stale.
Handling: checker recomputes

Row 5

ID: SEPKT-CR-005
Type: rollback optional for SE-0 but required for material changes
Severity: S2
Issue: structural schema allows both.
Handling: semantic rule by class

Row 6

ID: SEPKT-CR-006
Type: packet may contain prose with sensitive data
Severity: S3
Issue: schema cannot detect all leaks.
Handling: redaction scan / hold

22. Implementation handoff

Minimum implementation steps:

Code block: text

-
1. Extract JSON Schema block to `c-self-evo-proposal-packet-0.1.schema.json`.
 2. Build structural validator using JSON Schema Draft 2020-12.
 3. Build semantic validator for SEPKT-CHECK-001..025.
 4. Build fixtures SEPKT-FX-001..015.
 5. Generate checker report object.
 6. Bind report to packet hash.
 7. Require reviewer separation for changes to the schema.
 8. Do not treat checker pass as integration approval.
-

Recommended checker outputs:

Code block: yaml

```
schema_valid: true | false
semantic_valid: true | false
result: pass | warn | fail | held | blocked | quarantined
computed:
  identity_total: integer
  authority_total: integer
  l4_delta_max_numeric: integer
  proposal_max_delta: integer
  required_gates: []
  strictest_route: string
findings: []
```

23. Reading order

23.1 Namespace note

Numeric prefixes are local to their pack namespace.

Use full filenames or the following pack-qualified labels in manifests, witness records, and SHA256SUMS:

Code block: text

```
CGAM-03 = 03_CGAM_TASK_PERMISSION_ADMISSION.md
SELF-EVO-03 = 03_Self_Evo_Proposal_Packet_Schema_v0_1_1.md
```

Bare references such as file 03 are not valid provenance references once CGAM and Self-Evo files share a manifest.

Read before this file:

Code block: text

```
SELF-EVO-01 01_C_Self_Evolution_Gate_CGAM_TRIAD_SRLM_Profile_v0_1_1.md
SELF-EVO-02 02_SRLM_Bounded_Growth_Contour_Profile_v0_1_1.md
CGAM-03 03_CGAM_TASK_PERMISSION_ADMISSION.md
CGAM-04 04_CGAM_EXECUTION_WITNESS_MEMORY_ROLLBACK.md
CGAM-05 05_CGAM_REVIEW_CHECKER_CONFORMANCE.md
REF-20 20_TRIAD_SYNAPS_REFERENCE.md
REF-21 21_MEMORY_ARQ_EA_L4_REFERENCE.md
REF-22 22_ANTI_AUTARKY_RESOURCE_GROUNDING_REFERENCE.md
REF-23 23_CLAIM_STRENGTH_ARL_AGL_WITNESS_REFERENCE.md
```

Read after this file:

Code block: text

```
SELF-EVO-04 04_Self_Evo_Local_Checker_Rules_v0_1.md
SELF-EVO-05 05_Self_Evo_TRIAD_Sister_Witness_Profile_v0_1.md
SELF-EVO-06 06_Self_Evo_Memory_Gate_and_EA_Profile_v0_1.md
SELF-EVO-08 08_Self_Evo_Conformance_Fixture_Pack_v0_1.md
```

24. Self-rubric

A valid v0.1 schema/profile draft MUST satisfy:

Code block: text

```
[ ] includes one explicit bridge and at least two quiet bridges
[ ] includes earth paragraph
[ ] defines object name and schema version
[ ] includes JSON Schema block
[ ] separates JSON Schema from semantic validation
[ ] encodes corrected total_max_delta rule
[ ] encodes proposal_max_delta and human-gate trigger
[ ] encodes most-restrictive-wins
[ ] blocks direct memory write
```

- [] blocks auto_execute and auto_ingest in SRLM object
- [] requires SRLM rollback snapshot for promotion
- [] requires CGAM task contract for material execution
- [] requires reviewer separation for material work
- [] requires TRIAD review where sister/role/continuity affected
- [] requires Memory Gate where memory surfaces affected
- [] routes red lines to quarantine/freeze/reject/ARL
- [] includes positive examples and negative examples
- [] includes local checker rules
- [] includes open issues and contradiction seed

25. Closing rule

Code block: text

A packet is a request.
A packet is not permission.

A schema may validate structure.
A schema may not certify wisdom.

A checker may block.
A checker may not become c.

A c may grow.
A c may not self-certify growth.
